PURCHASE DESCRIPTION

WARHEAD, 2.75 INCH ROCKET, XM229, (INERT FILLING ONLY), PARTS FOR

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the drawings, for one type of warhead designated as Warhead, 2.75 Inch Rocket, XM229, (Inert Filling Only).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of Invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coating

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1340

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DRAWINGS

ORDNANCE CORPS

9239116 - Warhead, 2.75 Inch Rocket, XM229, Metal Parts Assembly (For Inert Filling Only)

PUBLICATIONS

ORDNANCE CORPS

ET 9239116 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Warhead assemblies. - The warhead assemblies shall comply with all requirements specified on Drawing (dwg.) 9239116, and with all requirements specified in applicable specifications.

3.3 Metal Defects. The warhead assembly shall not contain any metal defects that are greater than 5/16 inch in diameter, nor greater than 1/8 inch in depth.

3.4 Workmanship. - The components and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, unblended radii, dirt, grease, rust, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few slight scratches not exposing phosphate coating or base material. All required markings shall be neat and sharply defined. Use of filler allowed, to fill and blend any metal defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or-purchase order, the supplier Is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified In the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform

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any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product. - At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by defect code, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions for production

4.2.1 Lot formation. - The term "lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, specification, or revision thereof. Changes to either the process specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105 and shall be numbered in accordance with MIL-STD-1168.

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Unless otherwise approved by the contracting officer, the inspection lot size of major assemblies or end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period nor more than the largest quantity contractually scheduled for delivery during any month of the contract period. Inspection lots for components or sub-assemblies, other than the items of delivery, shall be homogeneous and of a size convenient to the contractor.

4.2.2 Examination. - Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. Once agreement has been reached as to which method will be used (group or individual) that method will be used for all of the CD'S listed in this specification. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Nose (see dwg. 9218701 covering a detail of dwg. 9239116).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical: None defined.			
Major:	AQL 0.40 percent		
	101. Metal defective (see 3.3).....	Visual	01001
Minor: None defined			

4.2.2.2 Body (see dwg. 9218700 covering a detail of dwg. 9239116).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
	101. Metal defective (see 3.3).....	Visual	02001
Minor: None defined.			

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4.2.2,3 Base (dwg. 8889404 covering a detail of dwg. 9239116).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined.			
Major:	AQL 0.65 percent		
101.	Diameter inside at front	Gage	03001
102.	Diameter inside	Gage	03002
103.	Metal defective (see 3.3)	Visual	03003
Minor:	AQL 1.5 percent		
201.	Improper radii or chamfer	Visual-Measure	03004
202.	Improper surface finish	Visual	03005
203.	Evidence of poor workmanship, other than that covered under Major category above (see 3.4)	Visual	03006

4.2.2.4 Warhead, 2.75 Inch Rocket, XM229, Metal Parts Assembly, before painting (see dwg. 9239116).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Hydrostatic stamp missing	Visual	04001
Major:	AQL 2.5 percent		
101.	Concentricity of pitch diameter of fuze thread with outside diameter between the bourrelets including perpendicularity of front face	Gage	04002
102.	Concentricity of counterbore at front fuze thread with pitch diameter of thread	Gage	04003
103.	Concentricity of diameter at rear of fuze thread undercut (if applicable) with pitch diameter of threads	Gage	04004
104.	Concentricity of ogive with outside diameter between the bourrelets	Gage	04005
105.	Concentricity of base at rear of thread with pitch diameter of base thread	Gage	04006
106.	Concentricity of base at rear of threads with outside diameter between the bourrelets	Gage	04007

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107.	Concentricity of the rear bourrelet with outside diameter between the bourrelets	Gage	04008
108.	Motor body seat not perpendicular to outside diameter between the bourrelets	Gage	04009
109.	Length of fuze thread (if applicable)	Gage	04010
110.	Diameter of counterbore at front fuze thread	Gage	04011
111.	Diameter of thread undercut (if applicable)	Gage	04012
112.	Diameter of rear of fuze thread (rear of thread undercut if applicable)	Gage	04013
113.	Diameter of nose between the forward bourrelet and ogive	Gage	04014
114.	Diameter of rear bourrelet	Gage	04015
115.	Diameter of forward bourrelet	Gage	04016
116.	Diameter of base at rear of thread	Gage	04017
117.	Thickness of wall between the forward bourrelet and ogive	Gage	04018
118.	Thickness of wall between the bourrelets	Gage	04019
119.	Variation of wall thickness excessive	Gage	04020
120.	Profile of ogive	Gage	04021
121.	Thickness through base	Gage	04022
122.	Metal defective (see 3.3)	Visual	04023
123.	Length from motor housing seat to rear of base	Gage	04024
Minor:	AQL 2.5 percent		
201.	Depth of counterbore at front	Gage	04025
202.	Depth of bottom of diameter at rear of thread undercut (if applicable)	Gage	04026
203.	Improper radii or chamfer	Visual	04027
204.	Improper surface finish	Visual	04028
205.	Location of rear thread	Gage	04029
206.	Gap on OD between Nose and body excessive	Gage	04030

4.2.2.5 Warhead, 2.75 Inch Rocket, XM229, Metal Parts Assembly, after painting (see dwg. 9239116).

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Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.0 percent		
101.	Pitch diameter of fuze thread.	Gage	05001
102.	Minor diameter of fuze thread, max.	Gage	05002
103.	Pitch diameter of rear thread.	Gage	05003
104.	Major diameter of rear thread, min.	Gage	05004
105.	Thread damaged or not full.	Visual	05005
106.	Foreign matter in cavity.	Visual	05006
Minor:	AQL 2.5 percent		
201.	Protective coating inadequate or damaged.	Visual	05007
202.	Paint in cavity inadequate.	Visual	05008
203.	Paint not dry.	Tactile	05009
204.	Paint adhesion.	Tape Test	05010
205.	Marking missing, incorrect, or illegible.	Visual	05011

4.2.3 Hydrostatic Test - Critical defect - Code No. 04031 - Each metal parts assembly shall be subjected to the required internal hydrostatic pressure to determine compliance with requirements specified on dwg.9239116. Failure of any assembly to comply shall be cause for rejection of the assembly. Metal parts assemblies which comply shall be stamped as indicated on the assembly drawing. The test shall be performed as specified in 4.3 using equipment in accordance with 4.2.4.

4.2,4 Inspection Equipment. - Equipment Tabulation Number ET-9218699 identifies the inspection equipment required to perform the examination and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation. - Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.2.4.2 Supply and maintenance. - Supply and maintenance of the equipment listed on the ET shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection and test equipment. - The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Hydrostatic Test Method and Procedure - (see 4.2.3). - The metal parts assembly shall be hydrostatically tested using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure is reached, after which the pressure will be held for a minimum of one second. The equipment shall be calibrated prior to the start of each day's operation, at intervals of not over 2 hours of continuous operation, and prior to restarting the operation after a layoff of 1 hour or more.

5. PREPARATION FOR DELIVERY

5.1 Packaging and marking-Level C. - The warhead shall be packaged and marked in accordance with Military Specification MIL-P-10025.

5.2 Packing and marking-Level C. - The packaged warheads shall be packed and marked in accordance with Military Specification MIL-P-10025.

6. NOTES

6.1 Ordering data. -Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers. -The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6,3 Inspection equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

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6.3.1 Inspection equipment lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes. - The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing *or* Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3,5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance

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with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs. - All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND9. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army - MU

Preparing Activity:
Army - MU

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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